PROJECT
(2020-2021)

COVID HELP
(CHAT BOT)

SYNOPSIS

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ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the synopsis of the B.Tech Mini Project **(Covid Help)** undertaken during B.Tech IIIrd Year. This project in itself is going to be an acknowledgement to the inspiration, drive and technical assistance will be contributed to it by many individuals.

We owe special debt of gratitude to **Mr. Piyush Vashistha,** Assistant Professor Department of CEA, for providing us with an encouraging platform to develop this project, which thus helped us in shaping our abilities towards a constructive goal and for his constant support and guidance to our work. His sincerity, thoroughness and perseverance is been a constant source of inspiration for us. We believe that he will shower us with all his extensively experienced ideas and insightful comments at different stages of the project & also taught us about the latest industry-oriented technologies.

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind guidance and co-operation.

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Covid Help

(Chat Bot)

It has been a whole year now since the COVID outbreak in 2020.All the countries went into lockdown , billions of lives lost , a number of small businesses were shut down and what not.
The virus containment situation was a bit better in india though thanks to our government. But still you can't keep a nation of 136.64 crores forever in lockdown now can you. So the lockdown was eventually lifted and the government started relaying instructions on how to keep yourself safe from the virus.

That's where Covid Help comes in. Covid Help is a python based chatbot that will mostly answer all your queries on novel coronavirus. Such as what is coronavirus, how to check if I have the symptoms for coronavirus or what even are the symptoms of coronavirus. Along with the coronavirus info the chatbot will also be able to answer basic conversation questions and some basic level math question.

But the sole main purpose of this bot is to help users with the info they need anytime without the hassle of searching for the information you need as the chatbot is providing you that same information.

So to summarize the chatbot will be able to answer the users coronavirus related queries and that's it.

TABLE OF CONTENTS

Declaration…… i

Certificate……. ii

Acknowledgement…… iii

Abstract……. iv

1. Introduction…..……... 1

1.1 Overview……... 1

1.2 Motivation……. 1

1.3 Problem Statement…… 1

1.4 Objective……... 2

2. Software Requirement Analysis……. 3

2.1 Dependencies/External Systems……... 5

2.2 Requirements…… 6

2.2.1 Functional Requirements……. 6

2.2.2 Non-Functional Requirements……. 6

2.2.3 External Interface Requirements…….. 7

3. Software Design…… 8

3.1 Data Flow Diagram……... 9

3.2 Use Case Diagram……. 12

4. Software Testing……... 15

4.1 Testing……... 15

4.2 Objectives of System Testing……... 15
4.3 Principles of System Testing…… 16
5. Implementation and User Interface……... 17
Bibliography and References…….. 20

PROJECT ANALYSIS:

Introduction

1.1 Overview:-

- All the functional/non-functional requirements, corresponding DFD's, UML and Use Case Diagrams have been organized in this report. Along with these designs, this report also contain the essential data of this project.
- The complete description of the application followed by the functionalities has been listed initially. Later on, the chat bot has been described diagrammatically with the help of different designing tools like Data Flow Diagram, Use Case Diagram, Interaction Diagram.

1.2 Motivation:-

In current scenario, as we all are facing an epidemic so it always helps to the greatest extent to be always informed of all the measures that we can take to keep others and ourselves safe. And to know everything that is necessary about the virus as well. So that we can plan or actions accordingly and keep ourselves and others safe.
As it can be hassle to search everything covid related on the internet yourself as you would have to go through dozens of websites looking for the info you want, I came up with the idea of this bot to help users query all the information they need to keep themselves safe from the virus and get knowledge about the virus as well.

1.3 Problem Statement:-

As we all are facing a pandemic right now so it is so it always helps to the greatest extent to be always informed of all the measures that we can take to keep others and ourselves safe. And to know everything that is necessary about the virus as well. So that we can plan or actions accordingly and keep ourselves and others safe.

I came up with the idea of this bot to help users query all the information they need to keep themselves safe from the virus and get knowledge about the virus as well.

1.4 Objective:

Covid Help aims to provide almost all the information about coronavirus to the user.

With **Covid Help** users can get to know all the information about coronavirus they need. Such was the scientific information about the virus, what to do if someone has is infected, what are the symptoms of the virus, how to check if someone has symptoms of the virus and so on, what are the government policies on fighting the virus and so on.

- There are following users: Anyone who has access to a smart device (such as a smart phone or a pc/laptop) with internet access & is interested in knowing the current status of the virus so far

- There are following user functionalities:

    i. Users can query almost all the information about coronavirus
    ii. Users can also query basic conversation questions with the bot

2. Dependencies/External Systems

User Interfaces:

The user interface for the chatbot is created with help of flask, html and CSS. The UI will include a text input area where user can input the question they want to ask and then bot will display the appropriate response on the message display area.

Hardware Interfaces:

The hardware interface will be the user's smart device and the server on which my bot is hosted.

HARDWARE REQUIREMENT (MINIMUM)

- 2 GB RAM
- **1 GB OF HDD SPACE**
- **Processor i5 (7th Gen)**
- **1024 x 768 Display**

SOFTWARE REQUIREMENT-

SYSTEM SOFTWARE

- **Operating System (Windows 10, Linux)**

APPLICATION SOFTWARE

- **Tools:**

PyCharm (any IDE or text editor would do) and several other python libraries which are listed below

- **Programming:**

PYTHON

Server:

- Heroku for hosting the bot

Technologies used:

Python: Python is the programming language of this project.

Python Libraries:

- **ChatterBot:** ChatterBot is a Python library that makes it easy to generate automated responses to a user's input. ChatterBot uses a selection of machine learning algorithms to produce different types of responses.

Flask: **Flask** is a web framework. This means **flask** provides you with tools, libraries and technologies that allow you to build a web application.

HTML: **HTML** (Hypertext Markup Language) is the code that is **used** to structure a web page and its content.

CSS: Cascading Style Sheets is a style sheet language used for describing the presentation of a document written in a markup language such as HTML.

References:

- www.w3school.com
- www.youtube.com